Arnav Gupta

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SUMMARY

Arnav Gupta is a highly accomplished high school student with a strong academic record and extensive experience in research, leadership, and extracurricular activities. He has demonstrated expertise in areas such as bioinformatics, chemistry, and biology, showcased through his research projects, academic achievements, and competitive participation. He also has leadership experience in various clubs and volunteering roles. In the future he would like to be a doctor.

EXPERIENCE

EXECUTIVE DEVELOPER

VOICES OF SOUTH ASIA(VOSA) October 2024 - present • Aggregated phonetic segment tokens across languages such as Hindi, Tamil, and Telugu, enhancing language model understanding of regional dialects using natural language processing techniques.

• Surveyed community institutions to encourage widespread adoption of curated Indian language token datasets, providing linguistic resources to improve LLM efficacy with computational linguistics skills.

• Leveraged email marketing software and analytics to disseminate the newsletter on a scheduled basis, monitoring engagement metrics to refine future editions.

STUDENT RESEARCHER

STANFORD UNIVERSITY July 2025 - present, Palo Alto, CA • Utilized QuPath software to analyze over 40 immunohistochemistry-stained liver tumor sections, applying trained tools for accurate identification of macrophages and T cells involved in collaborative research projects.

• Applied QuPath image analysis software to conduct comprehensive image analysis of immunohistochemistry-stained liver tumor sections, ensuring accurate quantification of cellular markers.

• Prepared comprehensive reports and visual data summaries of experimental findings using QuPath and data analysis techniques for review by research and clinical groups in the Dhanasekaran lab.

• Developed a comprehensive Standard Operating Procedure (SOP) outlining protocols for QuPath image analysis, providing clear guidance on handling liver tumor sections within the lab environment.

Lead Researcher

INDEPENDENT BIOINFORMATIC RESEARCH December 2024 - present, Fremont, CA • Utilized genetic datasets from a genome wide associated study and public clinical datasets to develop machine learning models predicting type 2 diabetes risk by integrating additional risk factors using Python and advanced data science techniques.

• Optimized machine learning algorithm training pipelines, evaluating contributory impact of individual risk variables through comprehensive modeling approaches with bioinformatics methodologies.

• Shared research findings on diabetes risk factor integration at the San Joaquin STEAM fair, facilitating exposure to current genetic and clinical data science techniques.

Student Bioinformatics Researcher

OXFORD UNIVERSITY July 2023 - November 2024 • Designed protocols for fine-tuning and validation of the ProtStab2 thermostability prediction model, incorporating data visualization with Gromacs to evaluate predicted protein structures.

• Executed controlled virtual experiments with ProtStab2 and visualized protein structures via Gromacs, applying molecular dynamics techniques to monitor conformational changes.

• Contributed to a joint bioinformatics project by sharing analysis insights with PhD fellows using Python and data science techniques.

High School Research Program

ASPIRED SCHOLARS DIRECTED RESEARCH PROGRAM November 2022 - August 2023, Fremont, CA • Performed experiments on graphene formation using Liquid Fold Exfoliation techniques to explore material properties for superconductor integration, leveraging laboratory protocols and Scanning Electron Microscope(SEM) imaging tools.

• Collaborated in a laboratory research environment to refine process parameters for graphene synthesis and SEM imaging, integrating Chemistry and microscopy laboratory protocols.

• Supervised and provided technical instruction to peers during Cyclic Voltammetry experiments to achieve reliable dataset generation. • Communicated research methodology and results for graphene synthesis and superconductor applications at the American Chemical Society (ACS) Sci-mix conference in San Francisco alongside peers and industry experts.

• Project awarded top honors in the Sci-mix interdivisional category at the American Chemical Society conference.

• Prepared poster for display and presentation at ASDRP symposium with over 1000 attendees.

EDUCATION

Honor Classes: Advanced Placement (AP) Biology, AP Calculus BC, AP Spanish Language, AP Statistics, AP Chemistry, AP World History, AP Psychology, Honors English 11, Honors Chemistry

MISSION SAN JOSE HIGH SCHOOL • Fremont, CA • 2022-2026 • 3.94/4.00

• Involvements: Model UN, Science Bowl, Speech and Debate, Science Olympiad.

AWARDS & HONORS

USA Biology Olympiad SEMIFINALIST

Center for Excellence in Education • 2024

• Most prestigious biology competition in the US.

• Placed in the top 150 high school students out of ~12,000 participants.

American Invitational Mathematical Examination(AIME) QUALIFIER

Mathematical Association of America • 2025

• Scored within the top percentile of AMC 10 and 12 competitors to qualify for the next stage in the series of prestigious high school mathematics competitions with over 50,000 participants.

BERKELEY CHEMISTRY TOURNAMENT

UC Berkeley • 2025

• Placed 3rd in individual division and 4th in polymers topic division at a chemistry tournament hosted at UC Berkeley.

NORCAL SCIENCE OLYMPIAD STATES TOURNAMENT

Northern California Science Olympiad • 2025

• Placed 5th overall in the Northern California Science Olympiad tournament out of 32 teams and previously placed 3rd in the Alameda county regionals qualifying tournament out of 39 teams.

PUBLICATIONS

Independent research project to predict thermal stability of I. Sakaiensis PETase in silico

SKILLS

Bioinformatics, Python, Gromacs, Computer Vision, Machine Learning, Data Analysis

Chemistry, Graphene Synthesis, Liquid Fold Exfoliation, Scanning Electron Microscope (SEM), Cyclic Voltammetry(CV), ACS

Biology, AP Biology, Biology Tutoring, Tutoring Fundraiser, USABO, Science Olympiad

Mathematics, AP Calculus BC, SAT, AMC 10, AMC 12, AIME, Berkeley Chemistry Tournament

Data Science, AI, LLMs, openGWAS, Clinical Data Analysis, Type 2 Diabetes Prediction